Tokenizer learning curve

Scholarly/code pretraining • 2026-10-16

18 of 18 requested candidates have measured Modal results. The grid doubles fitting text from 64 MiB through 2 GiB and compares 16,384, 32,768 and 50,000 vocabulary entries. All fresh candidates use the same source-weighted, parent-held-out evaluation corpus. A higher bytes-per-token value means fewer tokens for the same text; it does not establish better language-model quality.

Design: six fitting sizes × three vocabulary sizes, nested samples from 12 pinned sources, and a shared 32 MiB held-out evaluation corpus.

Metrics: bytes per token by source and overall; round trips and unknown tokens; vocabulary utilization and fixed-term fragmentation; fitting time and peak memory.

Interpretation: compression is a proxy. Evaluate source regressions and compute tradeoffs before choosing a tokenizer. No downstream model-quality result is implied.

Limitations: one sampling seed; no global fuzzy near-deduplication guarantee; cross-source parent reconciliation remains incomplete. The complete methodology and numeric comparisons are in REPORT.md and measurements.json.

Measured findings

The strongest observed compression is 4.1786 bytes/token at 2048 MiB and 50,000 entries. This identifies a compression result, not a final model-optimal vocabulary.

At 16,384 entries, 64 MiB → 2 GiB reduces held-out token count by 0.52%; the final doubling contributes 0.003%. The smallest sample within 1% of this vocabulary's best observed token count is 64 MiB. The 1% tolerance is an analysis heuristic.

At 32,768 entries, 64 MiB → 2 GiB reduces held-out token count by 0.81%; the final doubling contributes 0.017%. The smallest sample within 1% of this vocabulary's best observed token count is 64 MiB. The 1% tolerance is an analysis heuristic.

At 50,000 entries, 64 MiB → 2 GiB reduces held-out token count by 0.94%; the final doubling contributes 0.031%. The smallest sample within 1% of this vocabulary's best observed token count is 64 MiB. The 1% tolerance is an analysis heuristic.

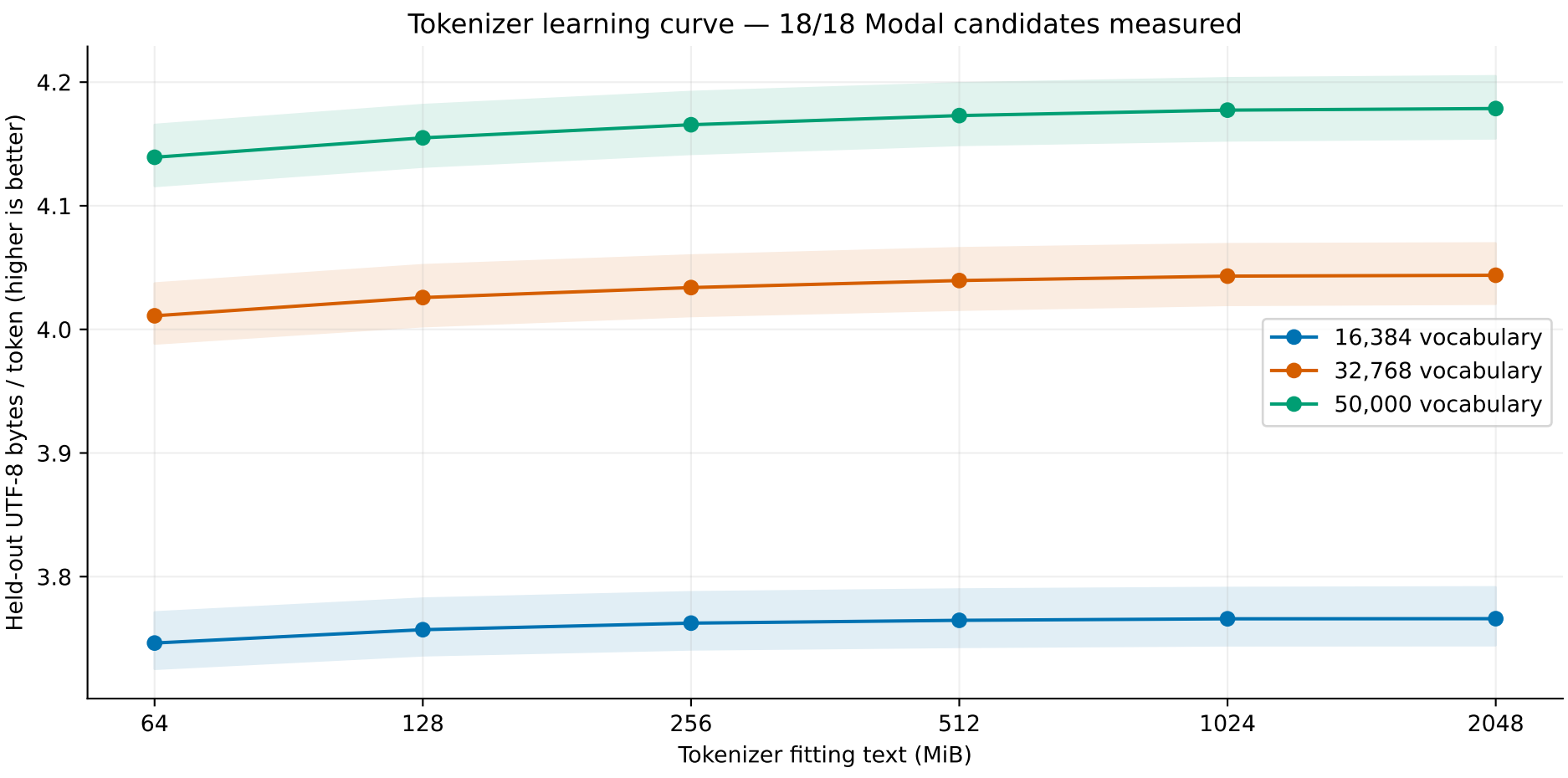
Measured findings (continued)

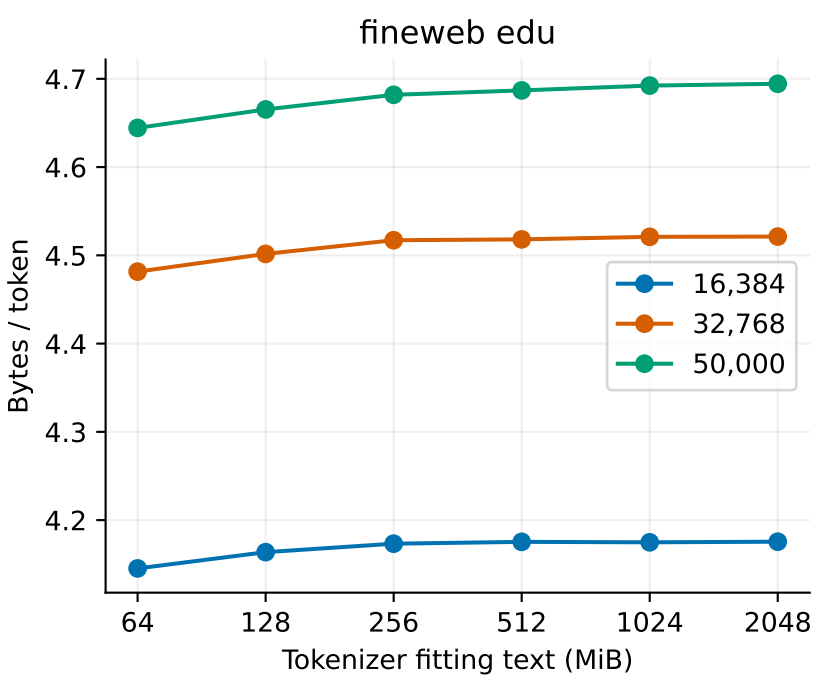
At 2 GiB, 32K uses 6.87% fewer tokens than 16K; 50K uses 9.88% fewer. At width 768, tied vocabulary parameters increase by 12.58M and 25.82M respectively. Untied input/output matrices double these deltas.

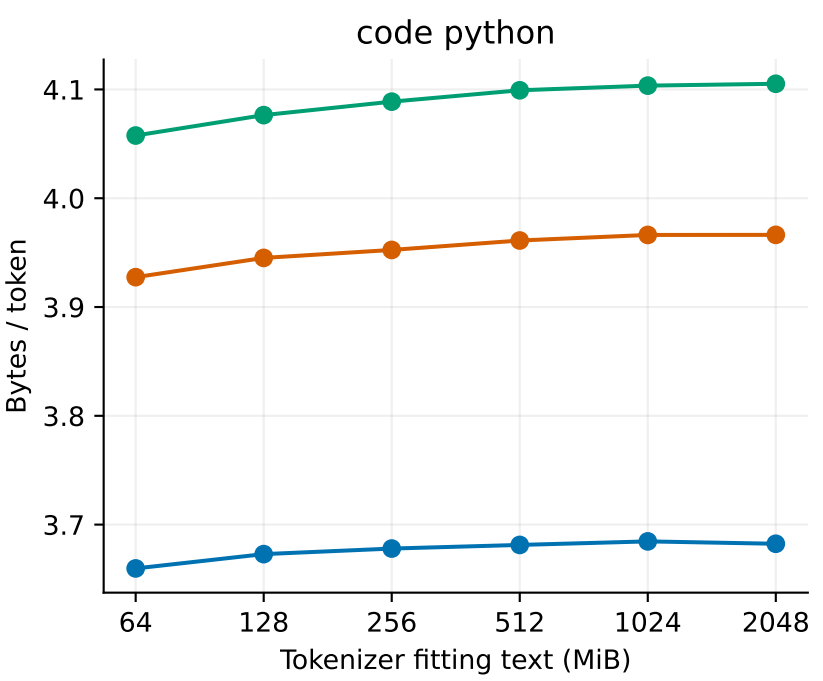
The published default is the 2 GiB / 32K candidate, a practical starting point balancing compression and vocabulary parameter overhead. All 18 variants are published for comparison. This choice has not been validated by matched-compute language-model training.

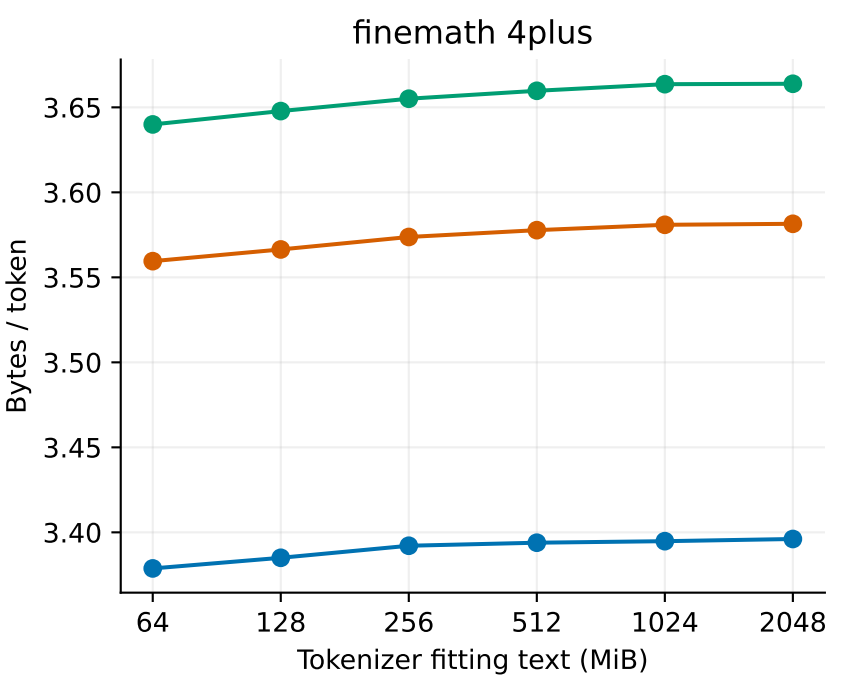
Frozen corpus composition — largest fitting sample and shared evaluation

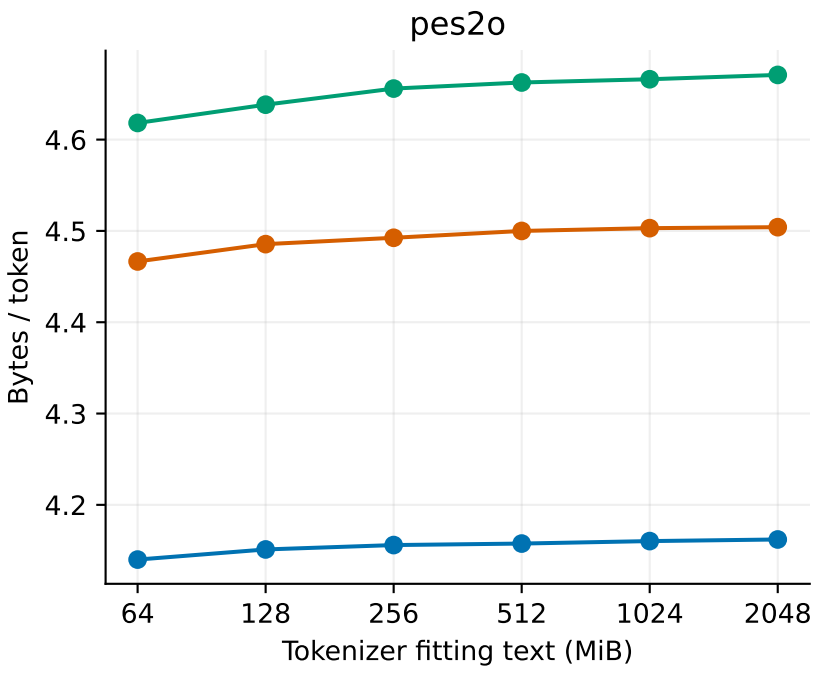
Source	Weight %	Fit MiB	Fit parents	Eval MiB	Eval parents
fineweb_edu	10	204.80	46886	3.20	845
code_python	30	614.40	39130	9.60	1394
finemath_4plus	20	409.60	87795	6.40	1573
pes2o	10	204.80	7302	3.20	214
pubmed	3	61.44	1689	0.96	66
arxiv	2	40.96	859	0.64	42
pressbooks	3	61.44	859	0.96	67
libretexts	3	61.44	367	0.96	67
doab	4	81.92	661	1.28	83
gutenberg	5	102.40	1886	1.60	108
pre_1929_books	7	143.36	2672	2.24	149
library_of_congress	3	61.44	1211	0.96	65

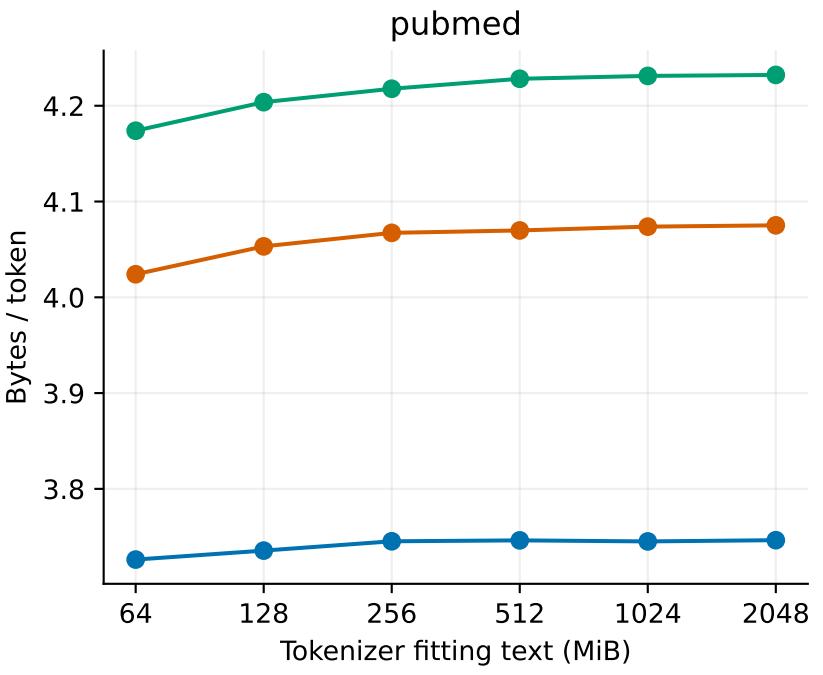

Tokenizer learning curve — 18/18 Modal candidates measured
Held-out UTF-8 bytes / token (higher is better)
4.2
4.1
4.0
3.9
3.8
64
128
256
512
1024
2048
Tokenizer fitting text (MiB)
16,384 vocabulary
32,768 vocabulary
50,000 vocabulary

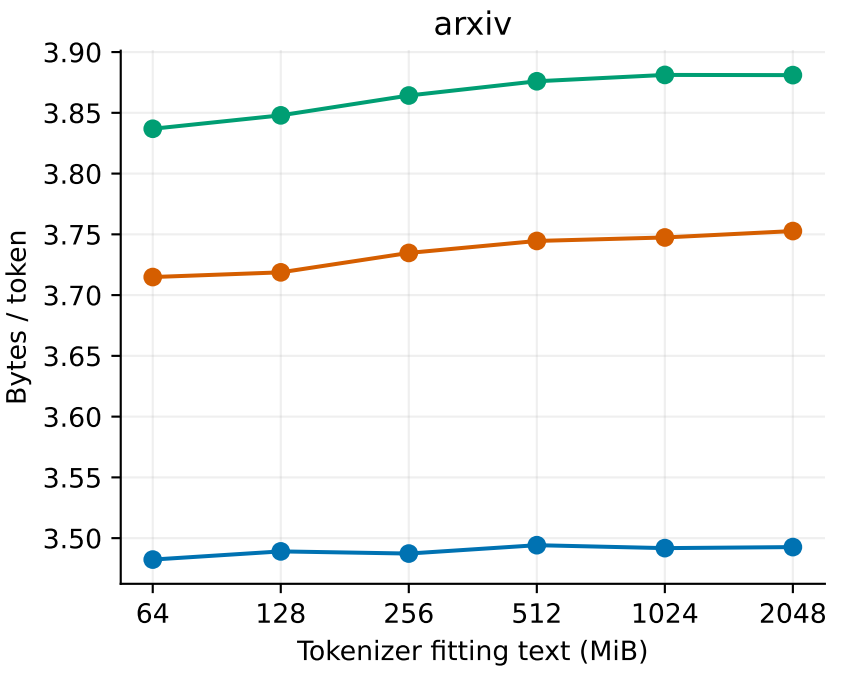

fineweb edu
Bytes / token
Tokenizer fitting text (MiB)
16,384
32,768
50,000

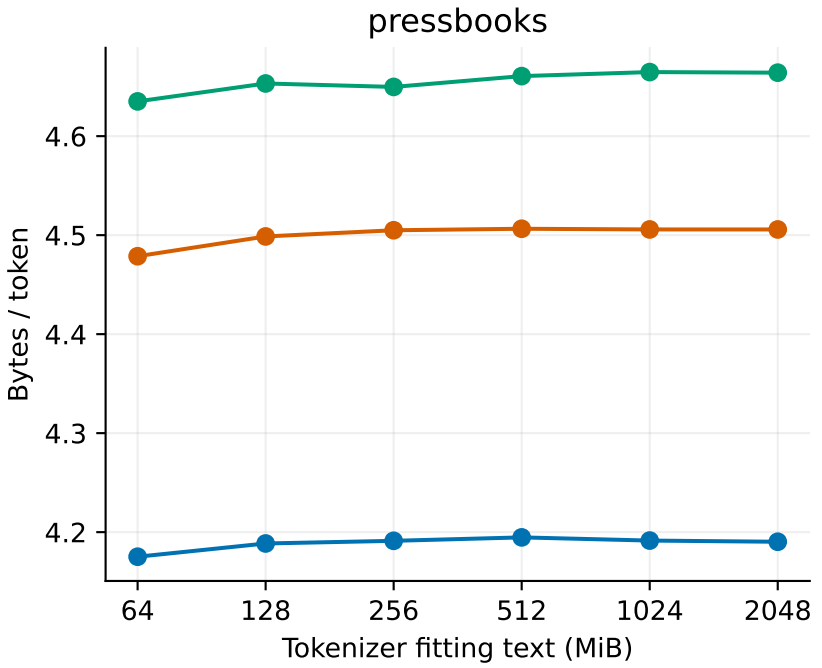

code python
Bytes / token
Tokenizer fitting text (MiB)

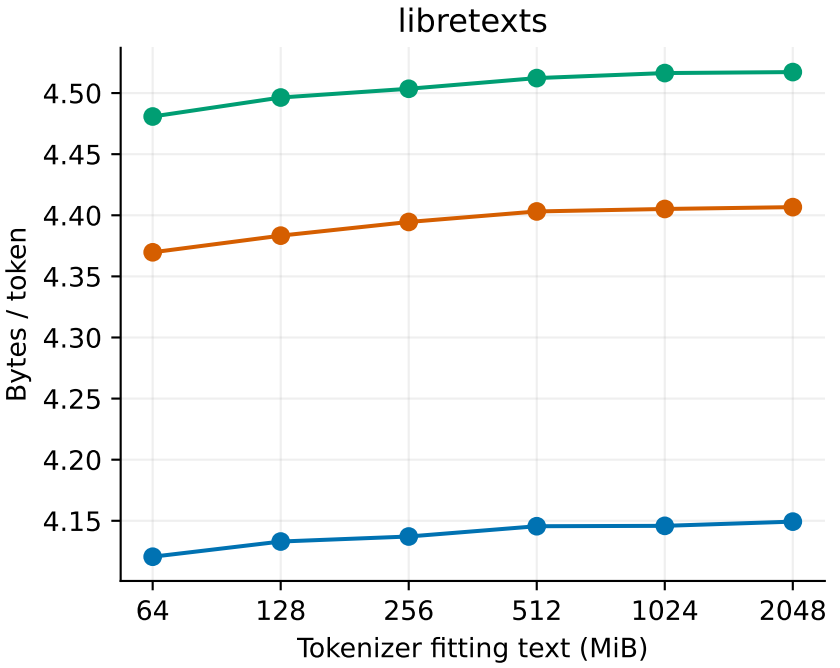

finemath 4plus
Bytes / token
Tokenizer fitting text (MiB)

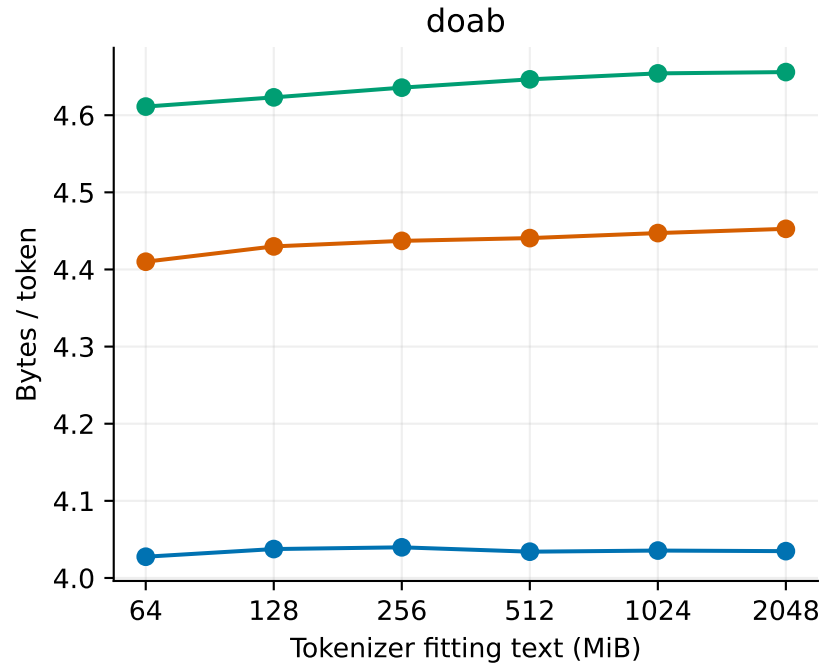

pes2o
Bytes / token
Tokenizer fitting text (MiB)

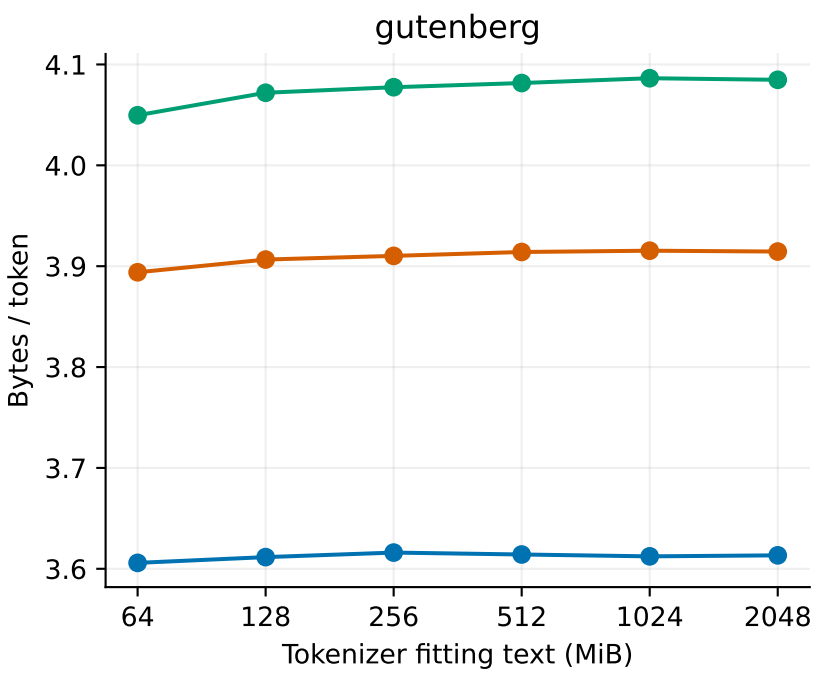

pubmed
Bytes / token
Tokenizer fitting text (MiB)

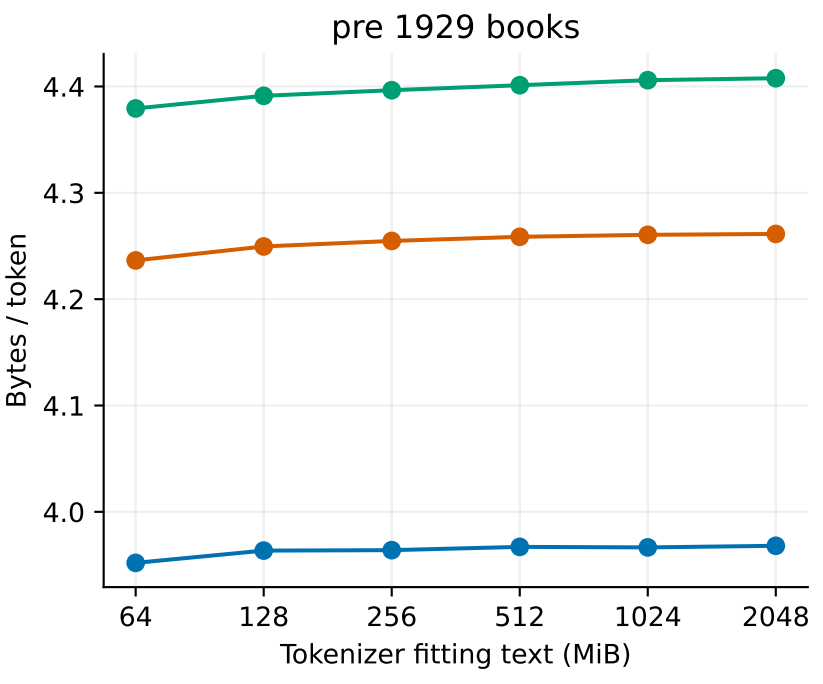

arxiv
Bytes / token
Tokenizer fitting text (MiB)

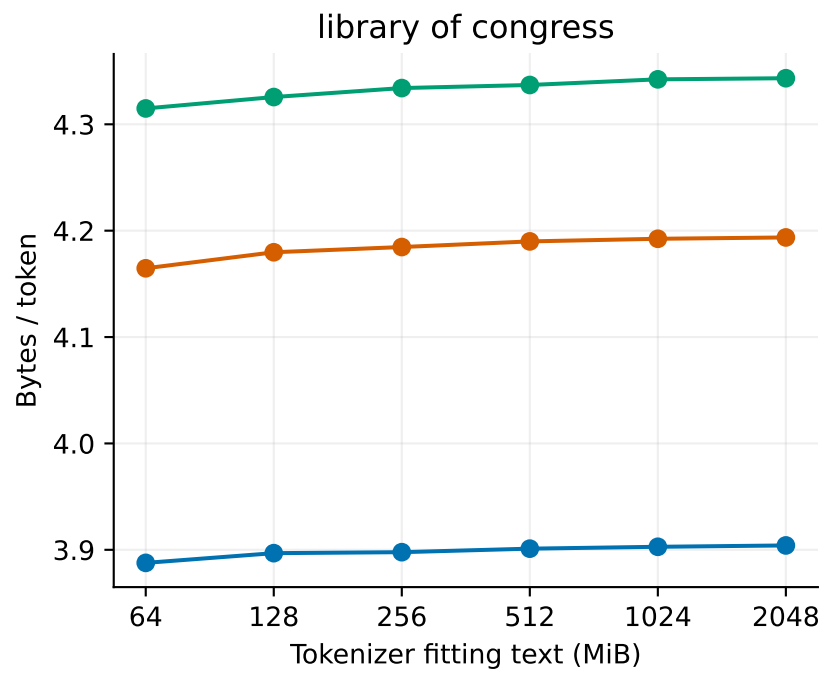

pressbooks
Bytes / token
Tokenizer fitting text (MiB)

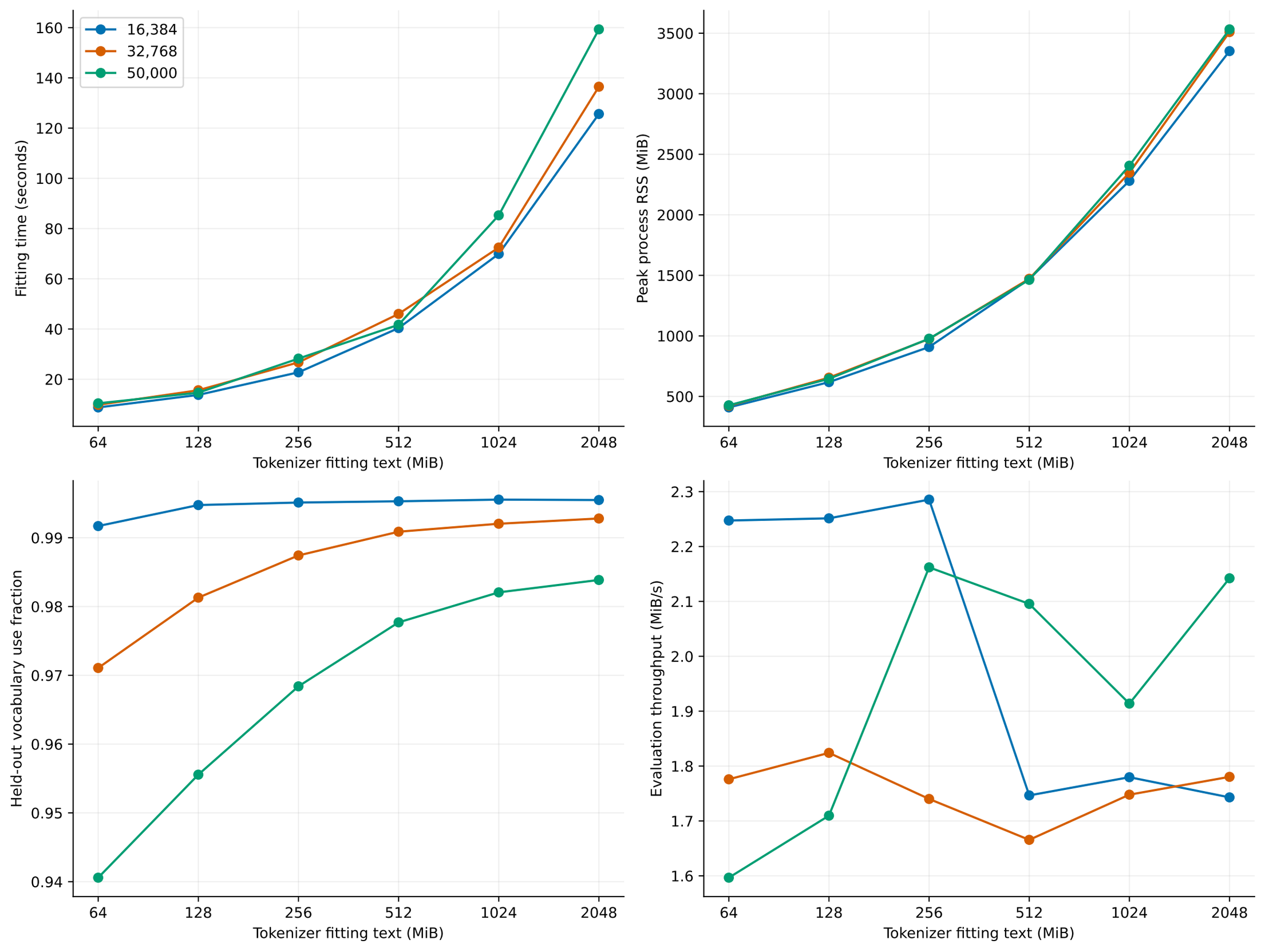

libretexts
Bytes / token
Tokenizer fitting text (MiB)

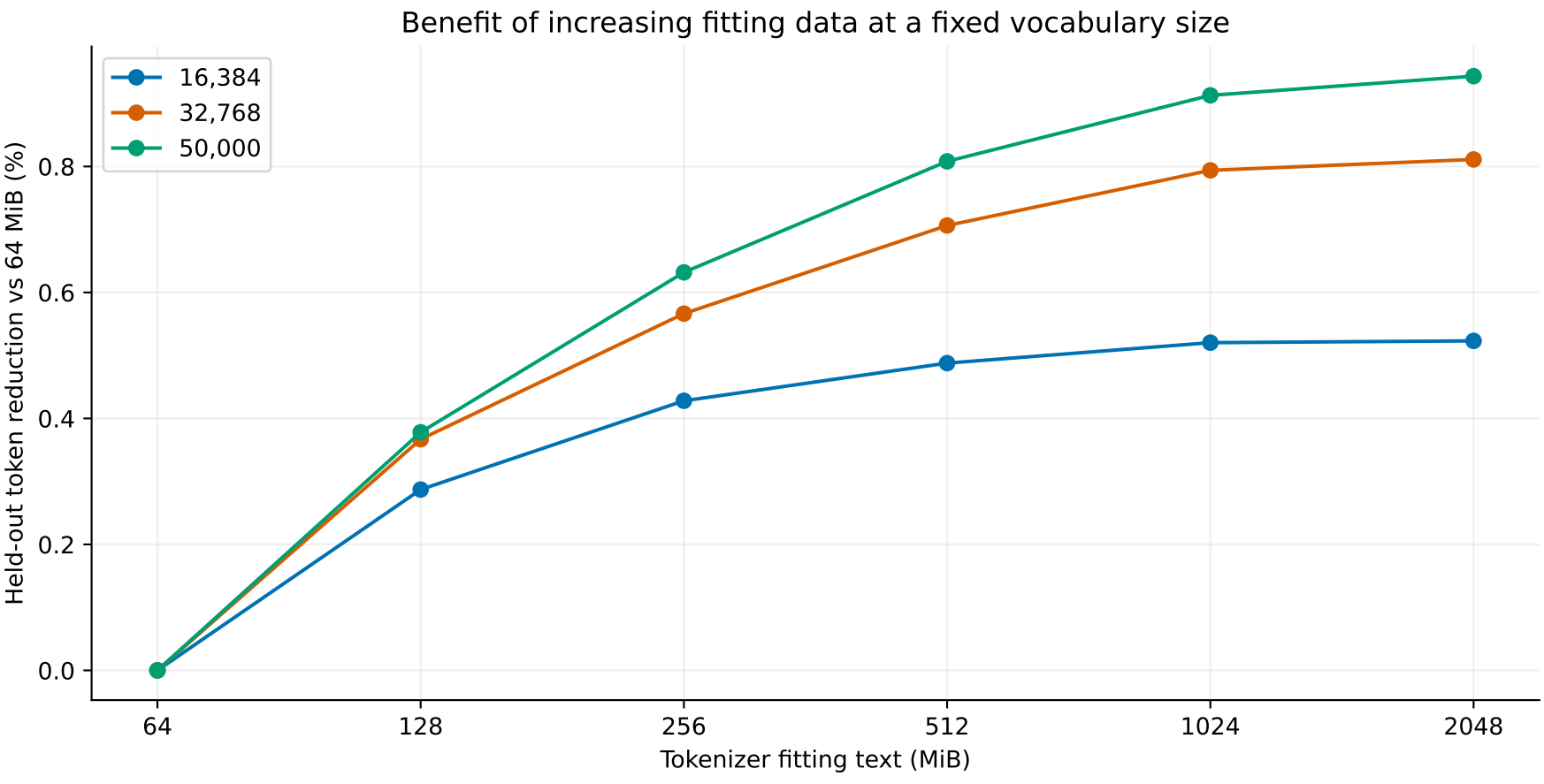

doab
Bytes / token
Tokenizer fitting text (MiB)

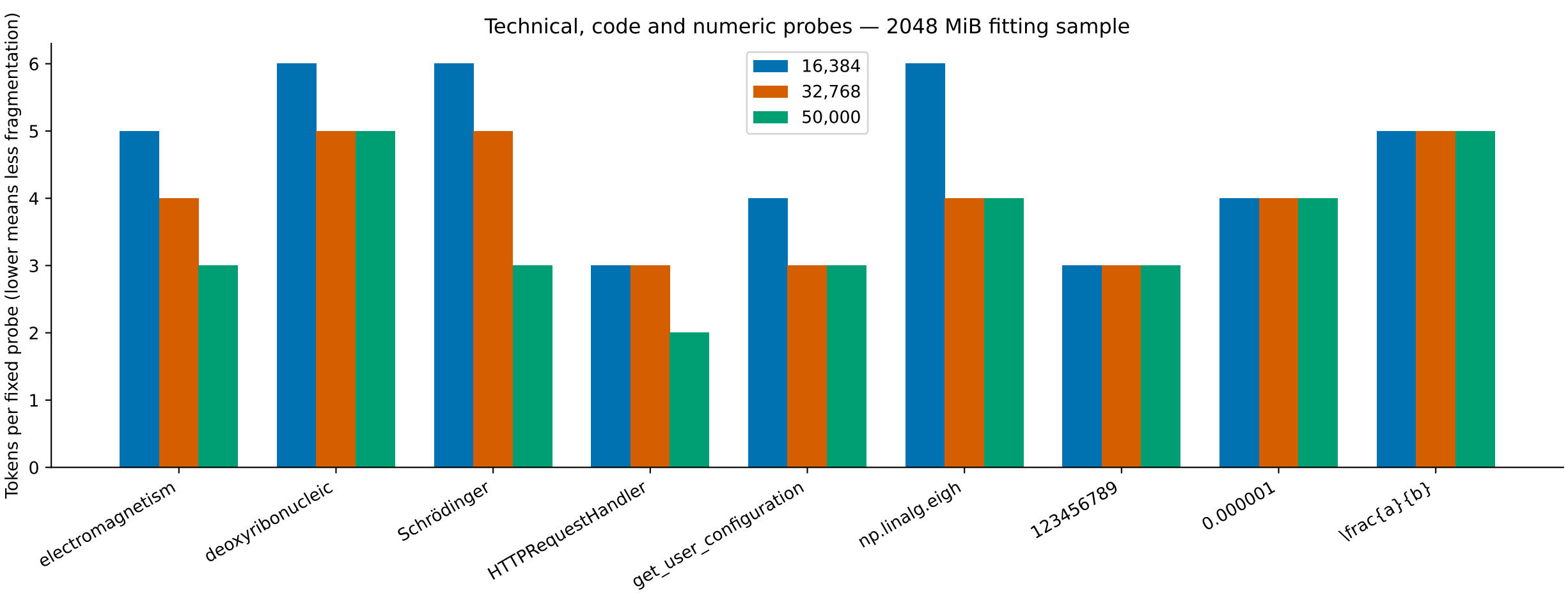

gutenberg
Bytes / token
Tokenizer fitting text (MiB)


pre 1929 books
Bytes / token
Tokenizer fitting text (MiB)


library of congress
Bytes / token
Tokenizer fitting text (MiB)


16,384
32,768
50,000
Fitting time (seconds)
Tokenizer fitting text (MiB)
Peak process RSS (MiB)
Tokenizer fitting text (MiB)
Held-out vocabulary use fraction
Tokenizer fitting text (MiB)
Evaluation throughput (MiB/s)
Tokenizer fitting text (MiB)
64
128
256
512
1024
2048


Benefit of increasing fitting data at a fixed vocabulary size
16,384
32,768
50,000
Held-out token reduction vs 64 MiB (%)
0.0
0.2
0.4
0.6
0.8
64
128
256
512
1024
2048
Tokenizer fitting text (MiB)


Technical, code and numeric probes — 2048 MiB fitting sample
Tokens per fixed probe (lower means less fragmentation)
16,384
32,768
50,000
0
1
2
3
4
5
6
electromagnetism
deoxyribonucleic
Schrödinger
HTTPRequestHandler
get_user_configuration
np.linalg.eigh
123456789
0.000001
\frac{a}{b}